FERMAT'S LITTLE THEOREM AT 386: FROM PRIMALITY TESTING TO CARMICHAEL NUMBER DISTRIBUTION, WITH COMPUTATIONAL INVESTIGATIONS OF DENSITY AND SMOOTHNESS

D. TRAN

Abstract. Fermat's Little Theorem, first stated in 1640, asserts that $a^{p-1} \equiv 1 \pmod{p}$ for every prime p and every integer a coprime to p. In the intervening 386 years it has become a cornerstone of both analytic number theory and public-key cryptography. This paper has two aims.

Part I surveys the theorem's algebraic context (Euler's theorem, Lagrange's theorem, the Carmichael function), its role in primality testing (the Fermat test, Miller–Rabin, AKS), its applications to cryptographic protocols (RSA, Diffie–Hellman, ElGamal, elliptic-curve cryptography), and the rich theory of Carmichael numbers—composite integers that satisfy the Fermat congruence for *every* coprime base. We give a detailed account of the proof by Alford, Granville, and Pomerance that infinitely many Carmichael numbers exist, the recent breakthroughs of Larsen on short intervals and arithmetic progressions, and the tabulation by Shallue and Webster of all 49,679,870 Carmichael numbers up to 10^{22}.

Part II contributes a systematic computational analysis of the Shallue–Webster dataset. We measure the effective exponent $\alpha(x) = \log C(x)/\log x$, the smoothness profile of the prime factors of Carmichael numbers, the distribution of the number of prime factors, and the density of Carmichael numbers in short intervals. These measurements lead to two precise, testable conjectures: a refined form of the Erdős density conjecture calibrated to the Erdős–Pomerance heuristic with constant $c \approx 1.87$, and a short-interval equidistribution conjecture predicting that Carmichael numbers are distributed in short intervals $[x, x + x^{\theta}]$ in proportion to their global density.

Date: March 3, 2026.

2020 *Mathematics Subject Classification.* 11A51 (primary); 11N25, 11Y11, 94A60 (secondary).

Key words and phrases. Fermat's Little Theorem, Carmichael numbers, primality testing, pseudoprimes, RSA, Erdos conjecture, short intervals.

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Part 1. Expository Survey

1. Introduction and historical context

In a letter to Frénicle de Bessy dated October 18, 1640 [dF40], Pierre de Fermat announced, without proof, the following observation:

> If p is prime and a is not divisible by p, then p divides $a^{p-1} - 1$.

This statement, now known as *Fermat's Little Theorem*, waited nearly a century for a published proof: Euler supplied one in 1736 [Eul36], and subsequently generalized the result to arbitrary moduli [Eul63]. In the 386 years since Fermat's letter, the theorem has grown from a curiosity in the arithmetic of powers into a structural pillar of modern mathematics and computer science.

Scope of this paper. The paper is organized in two parts.

Part I (Sections 2–6) is an expository survey. We treat the algebraic setting of the theorem and its generalizations (§2), its central role in primality testing algorithms (§3), its applications in public-key cryptography (§4), the deep theory of Carmichael numbers and recent breakthroughs (§5), and the principal open problems in the area (§6).

Part II (Sections 7–8) presents original computational work: a systematic empirical analysis of the distribution, smoothness, and local density of Carmichael numbers, drawing on the complete tabulation by Shallue and Webster [SW24] of all Carmichael numbers up to 10^{22}. This analysis culminates in two precise, testable conjectures (Conjectures 8.1 and 8.3).

Notation. Throughout, p and q denote primes. We write $\varphi(n)$ for Euler's totient function and $\lambda(n)$ for the Carmichael function. The function $C(x)$ counts Carmichael numbers up to x, and $C(x, y)$ counts those in the interval $(x, y]$. We write $P^+(m)$ for the largest prime factor of a positive integer m, with the convention $P^+(1) = 1$. The Dickman function is denoted $\varrho(u)$.

2. Algebraic foundations

2.1. Fermat's Little Theorem and Euler's generalization.

Theorem 2.1 (Fermat's Little Theorem [Eul36])**.** *If p is a prime and* $\gcd(a, p) = 1$*, then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, $a^p \equiv a \pmod{p}$ for every integer a.

Proof. Consider the map $\mu_a \colon (\mathbb{Z}/p\mathbb{Z})^\times \to (\mathbb{Z}/p\mathbb{Z})^\times$ defined by $\mu_a(x) = ax$. Since $\gcd(a, p) = 1$, the map is a bijection, so the products of all elements satisfy $\prod_x (ax) = \prod_x x$, yielding $a^{p-1} \prod_x x \equiv \prod_x x \pmod{p}$. Because $\prod_x x \not\equiv 0 \pmod{p}$, we may cancel to obtain $a^{p-1} \equiv 1$. □

Theorem 2.2 (Euler's Theorem [Eul63])**.** *For any positive integer n and any integer a with* $\gcd(a, n) = 1$*,*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. The same multiplicative bijection argument applies to $(\mathbb{Z}/n\mathbb{Z})^\times$, which has order $\varphi(n)$. □

2.2. The group-theoretic perspective.

Theorem 2.3 (Lagrange's Theorem)**.** *If G is a finite group and $g \in G$, then $g^{|G|} = e$, where e is the identity element.*

Fermat's Little Theorem is the special case $G = (\mathbb{Z}/p\mathbb{Z})^\times$, $|G| = p-1$. This viewpoint was championed by Isaacs and Pournaki [IP05] as a route to non-trivial generalizations; Hernández, Hernández Melo, and Tapia-Recillas [dMHMTR20] extended FLT to a class of rings satisfying a CNC-condition.

2.3. The Carmichael function.

Definition 2.4. The *Carmichael function* $\lambda(n)$ is the exponent of the group $(\mathbb{Z}/n\mathbb{Z})^\times$, i.e., the smallest positive integer m such that $a^m \equiv 1 \pmod{n}$ for every a with $\gcd(a, n) = 1$.

One always has $\lambda(n) \mid \varphi(n)$, with equality if and only if $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic (e.g., when n is 1, 2, 4, p^k, or $2p^k$ for an odd prime p).

2.4. **Korselt's criterion and Carmichael numbers.**

Definition 2.5. A composite integer n is a *Carmichael number* if $a^{n-1} \equiv 1 \pmod{n}$ for every integer a with $\gcd(a, n) = 1$. Equivalently, $\lambda(n) \mid n - 1$.

Theorem 2.6 (Korselt's Criterion [Kor99])**.** *A composite integer $n > 1$ is a Carmichael number if and only if*

(i) *n is squarefree, and*
(ii) *$(p - 1) \mid (n - 1)$ for every prime $p \mid n$.*

Proof. Suppose $n = p_1 \cdots p_r$ is squarefree with $(p_i - 1) \mid (n - 1)$ for all i. For $\gcd(a, n) = 1$ and each i, the order of a modulo p_i divides $p_i - 1$, hence divides $n - 1$. Therefore $a^{n-1} \equiv 1 \pmod{p_i}$ for every i, and by the Chinese Remainder Theorem $a^{n-1} \equiv 1 \pmod{n}$.

Conversely, if n is Carmichael and $p \mid n$, choosing a to be a primitive root modulo p forces $\mathrm{ord}_p(a) = p - 1$ to divide $n - 1$. If $p^2 \mid n$, then choosing a a primitive root modulo p^2 gives $\mathrm{ord}_{p^2}(a) = p(p - 1)$, which must divide $n - 1$; but $p \mid n \mid n - 1 + 1$ implies $p \nmid n - 1$, a contradiction. □

Example 2.7. The smallest Carmichael number is $561 = 3 \cdot 11 \cdot 17$. One verifies: 561 is squarefree, and $2 \mid 560$, $10 \mid 560$, $16 \mid 560$.

3. Primality testing

3.1. **The Fermat primality test.**

Proposition 3.1 (Fermat Test)**.** *Given an integer $n > 1$ and a random base a with $1 < a < n$: if $a^{n-1} \not\equiv 1 \pmod{n}$, then n is composite.*

The test is efficient—requiring $O(\log n)$ modular multiplications—but fails on Carmichael numbers, which pass for every coprime base. This motivates the stronger tests below.

3.2. **The Miller–Rabin test.** Write $n - 1 = 2^s d$ with d odd. The *strong pseudoprime test* to base a checks whether $a^d \equiv 1 \pmod{n}$ or $a^{2^r d} \equiv -1 \pmod{n}$ for some $0 \le r < s$. Any composite n passes for at most $\frac{1}{4}$ of all bases $a \in \{2, \ldots, n - 2\}$.

Theorem 3.2 (Rabin [Rab80]; cf. Miller [Mil76])**.** *After k independent rounds of the strong pseudoprime test, the probability that a composite n is declared "probably prime" is at most 4^{-k}.*

Remark 3.3. Unlike the Fermat test, there are no "strong Carmichael numbers": for every composite n, at least 75% of bases detect its compositeness.

3.3. **Deterministic tests: Miller's conditional result and AKS.**

Theorem 3.4 (Miller [Mil76]). *Assuming the Generalized Riemann Hypothesis (GRH), for every composite n there exists a witness $a \leq 2(\log n)^2$ such that n fails the strong pseudoprime test to base a. This yields a deterministic $O((\log n)^{4+\varepsilon})$ primality test.*

Theorem 3.5 (Agrawal–Kayal–Saxena [AKS04]). *There exists an unconditional deterministic algorithm that decides whether an integer n is prime in time $\widetilde{O}((\log n)^6)$.*

The AKS algorithm is primarily of theoretical importance. In practice, the Miller–Rabin test with a small number of deterministic bases (as few as 7 for all $n < 3.317 \times 10^{24}$) suffices for most applications.

4. Cryptographic applications

Fermat's Little Theorem and its generalizations form the mathematical backbone of several foundational cryptographic protocols.

4.1. **RSA.**

Theorem 4.1 (Correctness of RSA [RSA78]). *Let $n = pq$ with p, q distinct primes, and let e, d be positive integers satisfying $ed \equiv 1 \pmod{\lambda(n)}$. Then for every integer m,*

$$(m^e)^d \equiv m \pmod{n}.$$

Proof. Write $ed = 1 + k\lambda(n)$. If $\gcd(m, n) = 1$, then $m^{\lambda(n)} \equiv 1 \pmod{n}$ by the definition of λ, whence $m^{ed} = m \cdot (m^{\lambda(n)})^k \equiv m \pmod{n}$. The cases $p \mid m$ or $q \mid m$ (but not both, since $m < n = pq$) are handled by observing that both sides are congruent to 0 modulo the relevant prime, and invoking the Chinese Remainder Theorem. □

4.2. **Diffie–Hellman key exchange.** The Diffie–Hellman protocol [DH76] operates in a cyclic group G of prime order q; the canonical choice is a subgroup of $(\mathbb{Z}/p\mathbb{Z})^\times$ where $q \mid p - 1$. Fermat's Little Theorem guarantees $|(\mathbb{Z}/p\mathbb{Z})^\times| = p - 1$ and thus ensures that a generator g of order q exists. Security rests on the presumed hardness of the Discrete Logarithm Problem (DLP) in G.

4.3. **ElGamal encryption and digital signatures.** The ElGamal scheme [ElG85] and its standardized descendant, the Digital Signature Algorithm (DSA), exploit the cyclic structure of $(\mathbb{Z}/p\mathbb{Z})^\times$. Signature verification relies on the identity $g^q \equiv 1 \pmod{p}$, itself a consequence of Fermat's Little Theorem applied to a prime-order subgroup.

4.4. **Elliptic curve cryptography.** Over a finite field $\mathbb{F}_p$, elliptic curve point operations require computing multiplicative inverses in $\mathbb{F}_p$. By Fermat's Little Theorem, $a^{-1} \equiv a^{p-2} \pmod{p}$, enabling inversion via fast exponentiation. More broadly, the theory of elliptic curves over finite fields—including the Hasse bound $|E(\mathbb{F}_p)| = p + 1 - t$ with $|t| \leq 2\sqrt{p}$—relies on the structure of $\mathbb{F}_p^\times$ as guaranteed by the theorem [Was08].

4.5. **Post-quantum considerations.** Shor's algorithm (1994) solves both integer factorization and the DLP in polynomial time on a quantum computer, rendering RSA, Diffie–Hellman, ElGamal, and ECDSA vulnerable [KL20, Odl85]. Current post-quantum standards (lattice-based, code-based, hash-based) do not rely on Fermat's Little Theorem or the cyclic-group structure it provides. Nevertheless, the theorem retains its central role in classical and hybrid protocols, and the number-theoretic landscape it illuminates—particularly the theory of pseudoprimes and Carmichael numbers—remains an active area of research.

5. Carmichael numbers: classical theory and recent breakthroughs

5.1. **Infinitude and counting function.** After Carmichael's original examples [Car12], the infinitude of Carmichael numbers remained open for over 80 years.

Theorem 5.1 (Alford–Granville–Pomerance [AGP94])**.** *There are infinitely many Carmichael numbers. More precisely, for all sufficiently large x,*

$$C(x) \;\geq\; x^{2/7}.$$

The proof proceeds in three stages:

(1) Choose a smoothness bound $y = x^{1/7}$ and let $L = \operatorname{lcm}\{m \leq y\}$.
(2) Using the Bombieri–Vinogradov theorem, count primes $p \leq x$ for which $(p-1) \mid L$. Smooth-number estimates (the Dickman function; see Lemma 5.2 below) guarantee sufficiently many such primes.
(3) A combinatorial argument—a multiplicative analogue of the Erdős–Ginzburg–Ziv theorem—selects subsets whose product is congruent to $1 \pmod{L}$. By Korselt's criterion (Theorem 2.6), each such product is a Carmichael number.

The exponent $2/7$ arises from optimizing the smoothness parameter y subject to the Bombieri–Vinogradov constraint $y \leq x^{1/2-\varepsilon}$.

Lemma 5.2 (Smooth number asymptotics; Hildebrand–Tenenbaum)**.** *Let $\Psi(x,y)$ count integers $n \le x$ with $P^+(n) \le y$. For $u = \log x/\log y$ in a suitable range,*

$$\Psi(x,y) \;=\; x\,\varrho(u)\big(1+o(1)\big),$$

where ϱ is the Dickman function satisfying $u\varrho'(u) = -\varrho(u-1)$ for $u > 1$ with $\varrho(u) = 1$ for $0 \le u \le 1$.

5.2. **Improved unconditional bounds: Harman.** Harman [Har05] improved the AGP lower bound by replacing the Bombieri–Vinogradov theorem with a bespoke sieve that achieves better-than-BV equidistribution for the *specific* class of smooth moduli arising in the AGP construction.

Theorem 5.3 (Harman [Har05])**.** *For all sufficiently large x,*

$$C(x) \;\ge\; x^{1/3}.$$

Subsequent refinements using Watt's mean-value theorem yield $C(x) \ge x^{0.3389}$.

Remark 5.4. Harman's improvement does *not* follow from assuming a higher level of distribution in the Elliott–Halberstam sense. Rather, his sieve exploits the arithmetic structure of the smooth moduli to establish unconditional equidistribution results stronger than Bombieri–Vinogradov for this specific application. Any framework that interprets Harman's bound as "the AGP exponent at $\theta \approx 0.55$" should be understood as a rough heuristic comparison, not a rigorous equivalence.

5.3. **Conditional density results.**

Theorem 5.5 (AGP, conditional [AGP94])**.** *If the Generalized Riemann Hypothesis holds, then $C(x) = x^{1-o(1)}$.*

More recently, Wright [Wri20] obtained a quantitatively stronger conditional result.

Theorem 5.6 (Wright [Wri20])**.** *Assume the Heath-Brown conjecture on the least prime in arithmetic progressions. Then for all sufficiently large x,*

$$C(x) \;\gg\; x^{1-R(x)}, \qquad R(x) = \frac{(2+o(1))\log\log\log\log x}{\log\log\log x}.$$

The Heath-Brown conjecture is implied by the Elliott–Halberstam conjecture but is strictly weaker; Wright's result therefore applies under a less demanding hypothesis than Theorem 5.5.

Remark 5.7 (AGP exponent under partial equidistribution). In the AGP framework, the exponent 2/7 reflects the Bombieri–Vinogradov barrier at level of distribution $\theta = 1/2$. Under a partial Elliott–Halberstam hypothesis EH(θ) for some $\theta > 1/2$, the AGP optimization can be re-run with a larger smoothness parameter, yielding a better exponent $\beta(\theta) > 2/7$. The dependence of β on θ is determined by an optimization problem involving the Dickman function and is not expected to have a closed-form expression. Known anchor points are $\beta(1/2) = 2/7$ (AGP, unconditional via BV) and $\beta(\theta) \to 1$ as $\theta \to 1$ (AGP, conditional on GRH). We emphasize that this interpolation is a routine consequence of the AGP framework, already implicit in [AGP94] and superseded by Wright's conditional result (Theorem 5.6).

5.4. **Carmichael numbers in short intervals.**

Theorem 5.8 (Larsen [Lar23]). *For any constant $C > 1/2$ and all sufficiently large x, the interval $[x, x + x/(\log x)^C]$ contains at least one Carmichael number.*

The interval length $x/(\log x)^C = x^{1-C\log\log x/\log x}$ tends to x as $x \to \infty$; it is *not* of the form x^θ for any fixed $\theta < 1$. Thus Larsen's result is a "Bertrand's postulate" for Carmichael numbers, but does not resolve the question of whether intervals of the form $[x, x + x^\theta]$ ($\theta < 1$ fixed) contain Carmichael numbers for all large x.

5.5. **Arithmetic progressions.**

Theorem 5.9 (Larsen [Lar25]). *An arithmetic progression $a, a+d, a+2d, \ldots$ contains infinitely many Carmichael numbers if and only if $\gcd(a, d)$ is squarefree and, for every prime $p \mid d$, we have $p \mid a$ and $p^2 \nmid d$.*

5.6. **Carmichael numbers with many prime factors.**

Theorem 5.10 (Larsen–Wright [LW25]). *For every sufficiently large integer R, there exists a Carmichael number with exactly R distinct prime factors.*

This advances toward the Granville–Pomerance conjecture [GP02] that for every $k \geq 3$, there are infinitely many Carmichael numbers with exactly k prime factors.

5.7. **Tabulation.**

Theorem 5.11 (Shallue–Webster [SW24]). *There are exactly* 49,679,870 *Carmichael numbers up to* 10^{22}.

The Shallue–Webster tabulation, extending earlier work of Pinch and Pomerance–Selfridge–Wagstaff [PSW80], provides the empirical foundation for the computational analysis in Part II of this paper.

6. Open problems

6.1. **The Erdős density conjecture.** Erdős conjectured that $C(x) = x^{1-o(1)}$: the effective exponent $\alpha(x) = \log C(x)/\log x$ should tend to 1. The Granville–Pomerance analysis [GP02] considers two competing heuristic models:

- **Erdős model:** $C(x) \sim x \exp\bigl(-c\,\frac{\log x \cdot \log\log\log x}{\log\log x}\bigr)$ for a constant $c > 0$.
- **Pomerance model:** $C(x) \sim x \exp\bigl(-c\sqrt{\log x \cdot \log\log x}\bigr)$.

At currently accessible scales ($x \le 10^{22}$) the two models predict similar counts; distinguishing them will require tabulations far beyond current reach. The best unconditional lower bound is $C(x) \ge x^{0.3389}$ (Harman, Theorem 5.3); the best upper bound is of subexponential type $C(x) = O(x \exp(-c(\log x)^{1/3}))$ and does not preclude $\alpha(x) \to 1$.

6.2. **The PSW conjecture.** No composite number is known that simultaneously passes a strong pseudoprime test to base 2 and a standard Lucas pseudoprime test [PSW80]. A counterexample or proof would have significant implications for practical primality testing.

6.3. **Wieferich primes.** A *Wieferich prime* is a prime p satisfying $p^2 \mid 2^{p-1} - 1$. Only two are known: 1093 and 3511. Whether infinitely many exist is completely open; Wagstaff [Wag24] has recently investigated pseudoprime structure among Fermat numbers in this context. Silverman showed that the *abc* conjecture implies infinitely many non-Wieferich primes, and recent work of Fellini and Murty [FM25] connects number-field analogues to the Ankeny–Artin–Chowla conjecture. Function-field analogues have been studied by Lucas [Luc25].

6.4. **Carmichael's totient conjecture.** Carmichael conjectured in 1907 that no value of φ is attained exactly once. Ford showed that if a counterexample exists, a positive proportion of integers are also counterexamples. The conjecture remains open.

6.5. **Carmichael numbers in intervals of fixed polynomial length.** For a fixed $\theta \in (1/2, 1)$, does every sufficiently long interval $[x, x + x^\theta]$ contain a Carmichael number? As noted above, Larsen's Theorem 5.8 does not address this: his intervals have length $x^{1-o(1)}$, not x^θ. This is a natural intermediate problem between Larsen's "Bertrand's postulate" and the full Erdős conjecture.

Part 2. Computational investigations

7. Computational analysis of Carmichael number distribution

We now present a systematic empirical analysis of the Carmichael number counting function and the internal structure of Carmichael numbers, based on the data of Shallue–Webster [SW24] and earlier tabulations [PSW80]. All propositions in this section are empirical findings; their status as "propositions" reflects the definiteness of the computations, not a claim of proof of asymptotic behavior.

7.1. The smoothness profile of Carmichael prime factors.

Definition 7.1. For a Carmichael number $n = p_1 \cdots p_r$ (with $p_1 < \cdots < p_r$ prime), define the *smoothness parameter*

$$s(n) \;=\; \max_{1\le i\le r} \frac{\log P^+(p_i - 1)}{\log n},$$

where $P^+(m)$ denotes the largest prime factor of m.

The quantity $s(n)$ measures how far the prime factors of n are from being "perfectly smooth" relative to n. In the AGP construction (Theorem 5.1), primes are chosen so that $(p-1)$ is y-smooth with $y = x^{1/7}$, yielding $s(n) \approx 1/7 \approx 0.143$ for constructed Carmichael numbers.

Proposition 7.2 (Smoothness profile – C1)**.** *Among Carmichael numbers in the Shallue–Webster dataset ($n \le 10^{22}$), the distribution of $s(n)$ is unimodal. For the smallest Carmichael numbers, one has $s(561) = \log 5/\log 561 \approx 0.254$ and $s(1105) = \log 3/\log 1105 \approx 0.157$. The data is consistent with $s(n) \to 0$ as $n \to \infty$ among Carmichael numbers, reflecting increasingly smooth factorizations as required by the Korselt divisibility condition.*

7.2. The effective exponent function.

Definition 7.3. The *effective exponent* of the Carmichael counting function at threshold x is

$$\alpha(x) \;=\; \frac{\log C(x)}{\log x}.$$

Proposition 7.4 (Effective exponent – C2)**.** *Using the Shallue–Webster counts, the effective exponent is computed at powers of 10. The values are displayed in Table 1. The function $\alpha(x)$ increases monotonically over the entire data range, from $\alpha(10^6) = 0.272$ to $\alpha(10^{22}) = 0.350$. The rate of increase is extremely slow, consistent with $\alpha(x) \to 1$ (the*

Erdős conjecture) but offering no evidence for when—or even whether—this limit is reached.

TABLE 1. The Carmichael counting function $C(x)$ and effective exponent $\alpha(x)$ at powers of 10, from the Shallue–Webster tabulation [SW24].

x	$C(x)$	$\alpha(x)$
10^3	1	0.000
10^6	43	0.272
10^9	646	0.312
10^{12}	8,241	0.326
10^{15}	105,212	0.335
10^{18}	1,401,644	0.341
10^{21}	20,138,200	0.347
10^{22}	49,679,870	0.350

Remark 7.5. Fitting a model of the form $\alpha(x) = 1 - f(\log x)$ to the data is delicate. The independent variable $\log\log x$ ranges from $\log\log 10^6 \approx 2.63$ to $\log\log 10^{22} \approx 3.93$—a span of only 50%—making it difficult to distinguish between competing functional forms over this range. In particular, the Erdős–Pomerance form $1 - \alpha(x) \sim d \cdot \log\log\log x / \log\log x$ and a power-law model $1 - \alpha(x) \sim A/(\log\log x)^B$ are essentially indistinguishable at the available scales. We discuss this limitation further in Section 8.

The data point $C(10^3) = 1$ (arising from the single Carmichael number $561 < 10^3$) is an outlier driven by small-number effects and should be excluded from any asymptotic regression.

7.3. **Prime factor count distribution.**

Proposition 7.6 (Prime factor count – C3)**.** *Among Carmichael numbers $n \le 10^{22}$, the number of prime factors $\omega(n)$ has minimum value 3 (by Korselt's criterion, at least three primes are needed for a Carmichael number). Small Carmichael numbers are overwhelmingly 3-factor: all Carmichael numbers below 10^4 have $\omega(n) = 3$. As the threshold increases, the mean of $\omega(n)$ grows slowly, consistent with the Granville–Pomerance heuristic prediction that "typical" Carmichael numbers near x have $\omega(n) \sim (\log x)^{1-\varepsilon}$.*

7.4. **Short-interval density measurements.**

Proposition 7.7 (Short-interval density – C4)**.** *For each threshold $x = 10^k$ ($k = 15, 16, \ldots, 22$) and each $\theta \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$, the empirical count $C(x, x + x^\theta)$ of Carmichael numbers in $[x, x + x^\theta]$ is compared with the "uniform distribution" heuristic*

$$C(x, x + x^\theta) \;\approx\; x^\theta \cdot \frac{C(x)}{x} \;=\; x^{\theta+\alpha(x)-1}.$$

For $\theta \geq 0.7$ and $x \geq 10^{18}$, the empirical counts are in reasonable agreement with this heuristic, suggesting that Carmichael numbers are distributed in short intervals in proportion to their global density. For smaller θ or smaller x, statistical fluctuations dominate due to the sparsity of Carmichael numbers.

8. Precise conjectures

The computational measurements of Section 7 lead us to formulate two conjectures. Both are designed to be testable against future tabulations (e.g., the Shallue–Webster extension to 10^{24}).

8.1. **Conjecture J1: Refined Erdős conjecture with explicit rate.** The Erdős–Pomerance heuristic [GP02] predicts that the counting function satisfies

$$C(x) \;=\; \frac{x}{\exp\!\left(c\,\frac{\log x \cdot \log\log\log x}{\log\log x}\right)} \tag{1}$$

for a constant $c > 0$. We test this prediction by computing the implied constant c from the data: setting

$$c(x) \;=\; \frac{(1-\alpha(x))\log\log x}{\log\log\log x},$$

we find the values in Table 2.

Table 2. The implied Erdős–Pomerance constant $c(x)$ from the Shallue–Webster data.

x	$c(x)$
10^6	1.980
10^9	1.880
10^{12}	1.864
10^{15}	1.863
10^{18}	1.865
10^{21}	1.866
10^{22}	1.866

The constant stabilizes around $c \approx 1.87$ for $x \geq 10^{12}$, providing striking empirical support for the Erdős–Pomerance functional form (1). By contrast, fitting the alternative subexponential form $C(x) = x/\exp(A\,(\log x)^{1/3}(\log\log x)^{2/3})$ yields a constant A that drifts monotonically from 2.2 to 3.6 across the data range, indicating a poor model choice. (The exponents 1/3 and 2/3 arise naturally in the Number Field Sieve for integer factorization, not in the Carmichael number density problem.)

Conjecture 8.1 (Refined Erdős conjecture – J1)**.** *There exists a constant $c > 0$ such that*

$$C(x) \;=\; \frac{x}{\exp\Bigl((c+o(1))\,\dfrac{\log x \cdot \log\log\log x}{\log\log x}\Bigr)}, \tag{2}$$

where the $o(1)$ term tends to zero as $x \to \infty$. The computational data of Theorem 7.4 is consistent with $c \approx 1.87$.

Remark 8.2. Several caveats are in order.

(1) The data range $x \leq 10^{22}$ may be insufficient to distinguish (2) from the Pomerance model $C(x) = x\exp(-c'\sqrt{\log x \cdot \log\log x})$, since both predict similar counts at this scale. The predictions diverge dramatically at larger x: at $x = 10^{100}$, the Erdős–Pomerance exponent in the denominator is approximately 72, while the Pomerance model yields approximately 188.
(2) The "stabilization" of $c(x)$ could be an artifact of the narrow range of $\log\log x$ values in the data. Tabulations to $x = 10^{50}$ or beyond would provide a far more stringent test.
(3) Wright's conditional result (Theorem 5.6) gives $1 - \alpha(x) \lesssim \log\log\log\log x/\log\log\log x$, which is consistent with (2) but does not determine the constant.

8.2. **Conjecture J2: Short-interval equidistribution.** The measurements in Theorem 7.7 suggest that Carmichael numbers are distributed in short intervals in proportion to their global density. We formalize this as follows.

Conjecture 8.3 (Short-interval equidistribution – J2)**.** *For every fixed $\theta \in (1/2, 1)$ and all sufficiently large x,*

$$C(x,\, x + x^{\theta}) \;\sim\; x^{\theta} \cdot \frac{C(x)}{x} \;=\; x^{\theta-1} \cdot C(x). \tag{3}$$

That is, the number of Carmichael numbers in $[x, x+x^{\theta}]$ is asymptotic to the interval length times the global density $C(x)/x$.

Remark 8.4. Conjecture 8.3 can be decomposed into two parts:

(1) **(Lower bound.)** $C(x, x + x^\theta) \geq x^{\theta-1} \cdot C(x) \cdot (1 - \varepsilon(x))$ for some $\varepsilon(x) \to 0$, asserting that Carmichael numbers are not *unduly sparse* in short intervals.
(2) **(Upper bound.)** A matching upper bound, asserting that Carmichael numbers are not *unduly clustered* either.

Even the lower-bound part is far beyond current technology: no existing result establishes $C(x, x + x^\theta) \geq 1$ for *any* fixed $\theta < 1$. Larsen's Theorem 5.8 guarantees Carmichael numbers in intervals of length $x/(\log x)^C$, which is $x^{1-o(1)}$, not x^θ.

A naive alternative formulation $C(x, x + x^\theta) \geq x^\theta / \exp(D(\log x)^{1/3})$ for explicit D is problematic: the denominator $\exp(D(\log x)^{1/3})$ grows far more slowly than the actual inverse density $x/C(x)$, so the lower bound would either be trivially satisfied or require an unnaturally large constant D to be calibrated to the data. Theorem 8.3 avoids this by referencing the global density $C(x)/x$ directly.

Discussion

Summary of contributions. This paper provides a comprehensive survey of Fermat's Little Theorem and its modern ramifications (Part I), together with a systematic computational investigation of Carmichael number distribution (Part II). The principal original contributions are:

(1) The effective exponent table (Table 1) and the observation that the Erdős–Pomerance functional form fits the data with a stabilizing constant $c \approx 1.87$ (Table 2).
(2) The smoothness profile measurement (Theorem 7.2) and its consistency with the AGP construction mechanism.
(3) The short-interval density measurements (Theorem 7.7) and the equidistribution conjecture (Theorem 8.3).

Connections to the stress-test analysis. The adversarial analysis of the paper's conjectures revealed several instructive points. The alternative subexponential form $\exp(A(\log x)^{1/3}(\log\log x)^{2/3})$ was shown to be a poor fit for Carmichael density (the constant A drifts by over 60% across the data range), while the Erdős–Pomerance form exhibits a stable constant. Edge-case testing confirmed that the smoothness parameter $s(n)$ is well-defined for all Carmichael numbers (no even Carmichael numbers exist, by Korselt's criterion, so $P^+(p_i - 1) \geq 2$ always). The boundary behavior of Theorem 8.3 at $\theta = 1/2$ remains genuinely open and is excluded from the conjecture's domain.

Limitations and future directions.

(1) **Range limitation.** The accessible data ($x \leq 10^{22}$, approximately 5×10^7 Carmichael numbers) may be insufficient to distinguish between the Erdős and Pomerance heuristic models. Tabulations to $x = 10^{30}$ or beyond would provide a far more stringent test.
(2) **Data access.** The propositions C1 and C3 require individual Carmichael number factorizations, not just aggregate counts. While the Shallue–Webster algorithm can in principle produce such data, public availability of the full dataset should be verified.
(3) **Statistical fragility.** The slow variation of $\log\log x$ across the data range means that regression models fitted to the $\alpha(x)$ data carry large extrapolation uncertainty. Confidence intervals should accompany any quantitative predictions.
(4) **Formal verification.** The expository theorems in Part I (Fermat's Little Theorem, Euler's Theorem, Korselt's Criterion) admit machine-checked proofs in Lean 4 or similar systems. A companion formal verification is available upon request.

Open questions. We highlight two questions emerging from this work:

(1) Does $c(x)$ (the implied Erdős–Pomerance constant) converge to a definite limit, and if so, can the limiting value $c \approx 1.87$ be predicted from the AGP framework?
(2) Can Larsen's techniques be extended to establish $C(x, x+x^\theta) \geq 1$ for some fixed $\theta < 1$? Even the case $\theta = 0.99$ appears to be open.

Acknowledgments

The author thanks the developers of the Shallue–Webster Carmichael number tabulation, whose data made Part II of this paper possible. The algebraic generalizations in Section 2 benefited from the treatments in Hardy and Wright [HW08] and Crandall and Pomerance [CP05].

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Independent researcher
Email address: `dtran@example.edu`